Versorgung

Versorgung: 1S-Pfad (Elektronik, Lader) und 2S-Pfad (Aktoren, Pyro), Lastschalter

Datei: power.kicad_sch

Teensy

Teensy 4.1, CS-Decoder und Analogmux, Stapelstecker J_STACK

Datei: mcu.kicad_sch

Sensorik

IMU ICM-42688-P und zwei BMP390 an SPI1, Qwiic an I2C

Datei: sensors.kicad_sch

Funk

LoRa E22-400T22S an Serial8, GPS an Serial1

Datei: radio.kicad_sch

Aktoren

Vier Servos und Rollmotor DRV8833 am 2S-Akku

Datei: actuators.kicad_sch

Pyro

Vier Pyro-Kanaele low-side (3 und 4 DNP), Arming, Durchgangspruefung

Datei: pyro.kicad_sch

Signal

WS2812B-Kette mit Pegelwandler, Summer, Kamera-Trigger

Datei: signal.kicad_sch

Kartonaut – Flugcomputer-Mainboard rev B (54 x 120 mm, 4 Lagen, JLCPCB)

Spezifikation: hardware/mainboard/README.md; Design Guide: docs/DesignGuide_Mainboard_revB.md

Massen: GND = Logik und 1S-Minus; GND_PWR = Rueckstrom aller 2S-Verbraucher. Einzige Verbindung: NT1 am 2S-Minus.

Dieser Schaltplan wird von gen_schematic.py aus circuit.py erzeugt. Aenderungen dort machen, dann neu erzeugen.

Teensy 4.1, ICM-42688-P, BMP390, E22-400T22S, 1S + 2S

Kartonaut

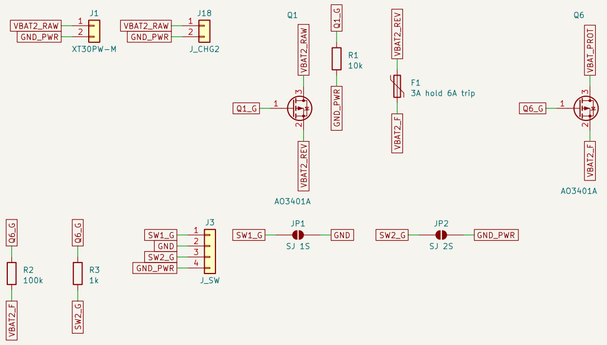
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File: mainboard.kicad_sch

Title: Mainboard rev B

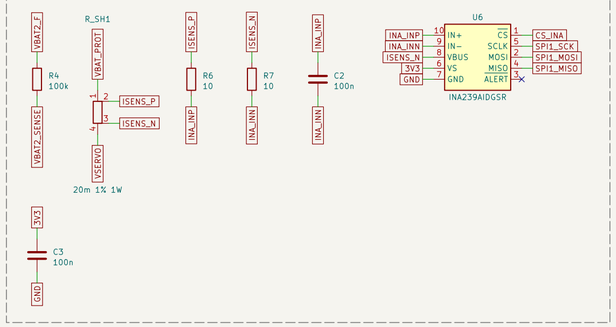
Size: A3	Date: 2026-10-21	**Rev: B**
KiCad E.D.A. 10.0.6		Id: 1/8

Versorgung: 1S–Pfad (Elektronik, Lader) und 2S–Pfad (Aktoren, Pyro), Lastschalter

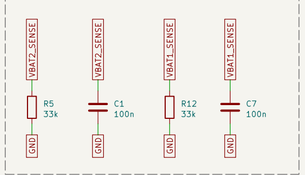
2S: Akku, Verpolschutz, Sicherung, Lastschalter



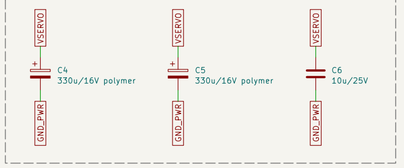
2S: Spannungsmessung, Shunt und INA239



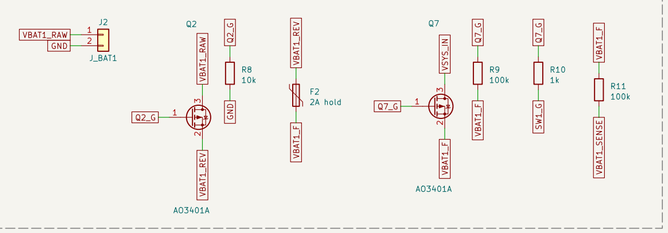
Messteiler am Analogmux



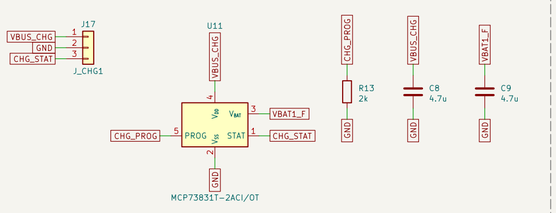
2S: Servo–Schiene VSERVO



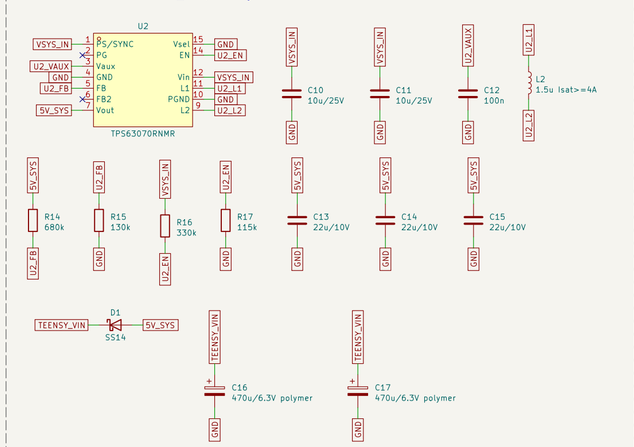
1S: Akku, Verpolschutz, Sicherung, Lastschalter



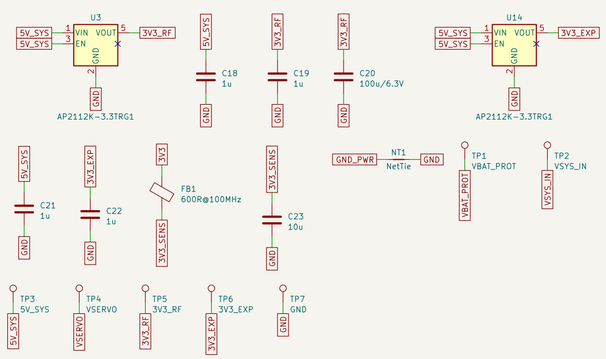
1S: Lader MCP73831



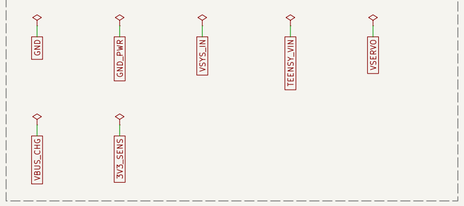
1S: Buck–Boost U2 –> 5V_SYS, Teensy–VIN



LDOs 3V3_RF und 3V3_EXP, Sensorinsel, Sternpunkt



Power–Flags fuer den ERC



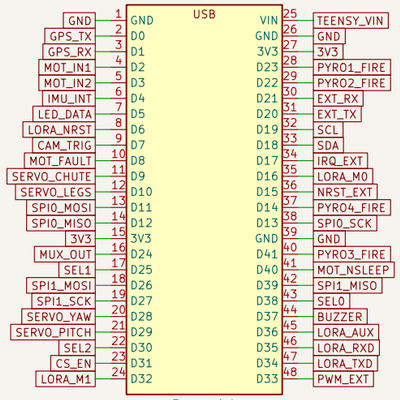
erzeugt aus circuit.py (gen_schematic.py)
Versorgung: 1S–Pfad (Elektronik, Lader) und 2S–Pfad (Aktoren, Pyro), Lastschalter
Kartonaut
Sheet: /Versorgung/
File: power.kicad_sch

Title: Mainboard rev B – Versorgung

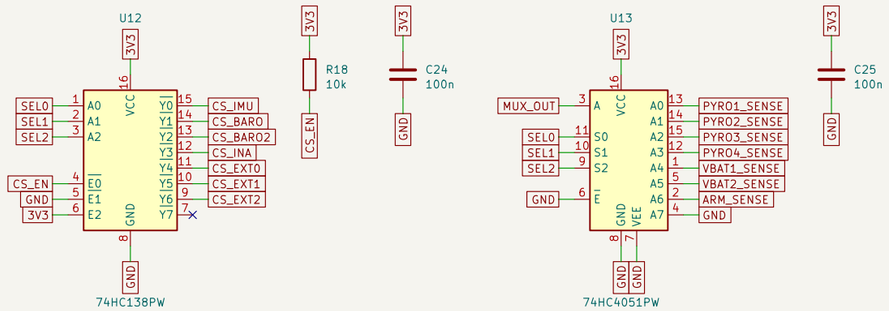
Size: A2	Date: 2026-10-21	**Rev: B**
KiCad E.D.A. 10.0.6		Id: 2/8

Teensy 4.1, CS–Decoder und Analogmux, Stapelstecker J_STACK

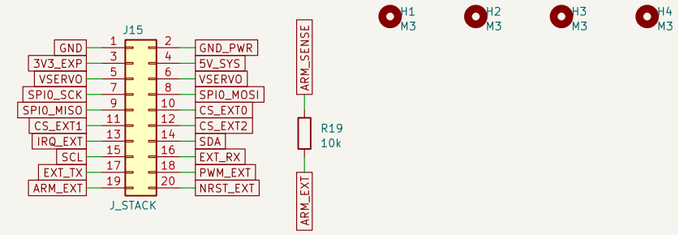
Teensy 4.1



CS–Decoder und Analogmux



Stapelstecker J_STACK, Befestigung, Testpunkte



erzeugt aus circuit.py (gen_schematic.py)
Teensy 4.1, CS–Decoder und Analogmux, Stapelstecker J_STACK
Kartonaut

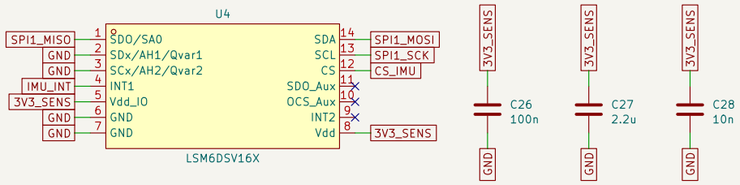
Sheet: /Teensy/
File: mcu.kicad_sch

Title: Mainboard rev B – Teensy

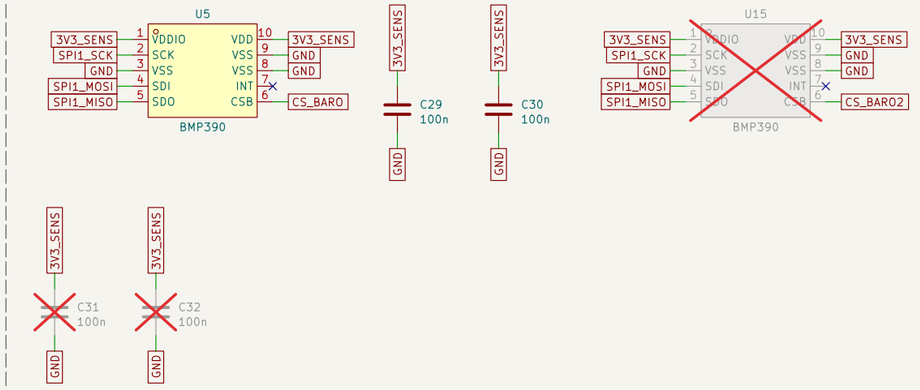
Size: A3	Date: 2026–10–21	**Rev: B**
KiCad E.D.A. 10.0.6		Id: 3/8

IMU ICM-42688-P und zwei BMP390 an SPI1, Qwiic an I2C

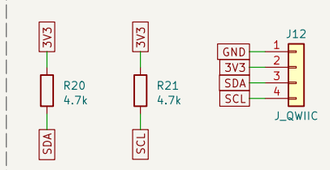
IMU ICM-42688-P, SPI1



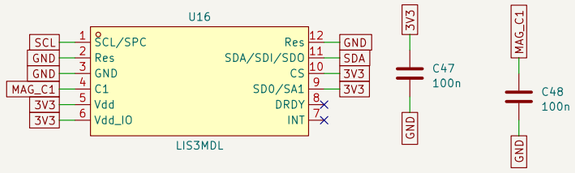
Barometer BMP390, SPI1



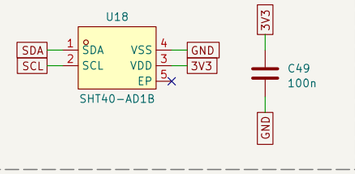
Qwiic, I2C



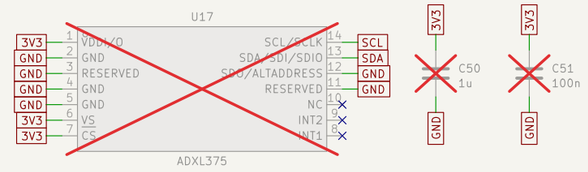
Magnetometer LIS3MDL, I2C (23.10.)



Feuchte und Temperatur SHT40, I2C (23.10.)



Hochbeschleunigung ADXL375, I2C, DNP (23.10.)



erzeugt aus circuit.py (gen_schematic.py)
IMU ICM-42688-P und zwei BMP390 an SPI1, Qwiic an I2C

Kartonaut

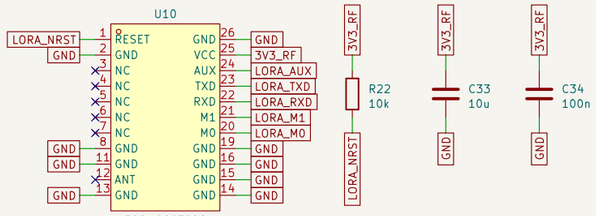
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File: sensors.kicad_sch

Title: Mainboard rev B – Sensorik

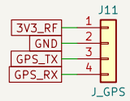
Size: A3	Date: 2026-10-21	**Rev: B**
KiCad E.D.A. 10.0.6		Id: 4/8

LoRa E22-400T22S an Serial8, GPS an Serial1

LoRa E22-400T22S, Serial8



GPS, Serial1



erzeugt aus circuit.py (gen_schematic.py)
LoRa E22-400T22S an Serial8, GPS an Serial1
Kartonaut

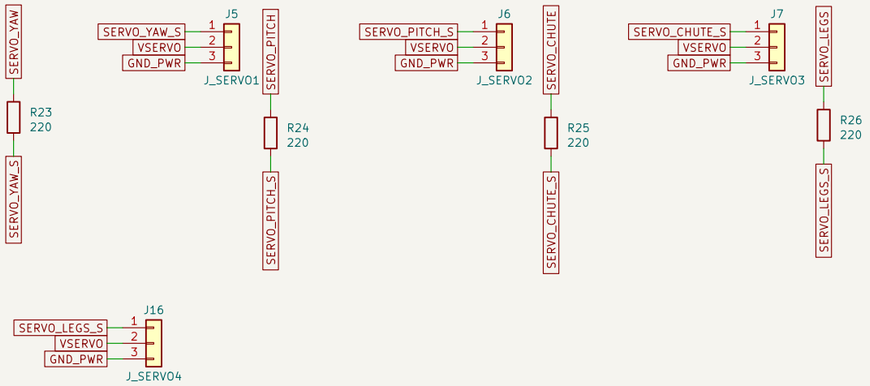
Sheet: /Funk/
File: radio.kicad_sch

Title: Mainboard rev B – Funk

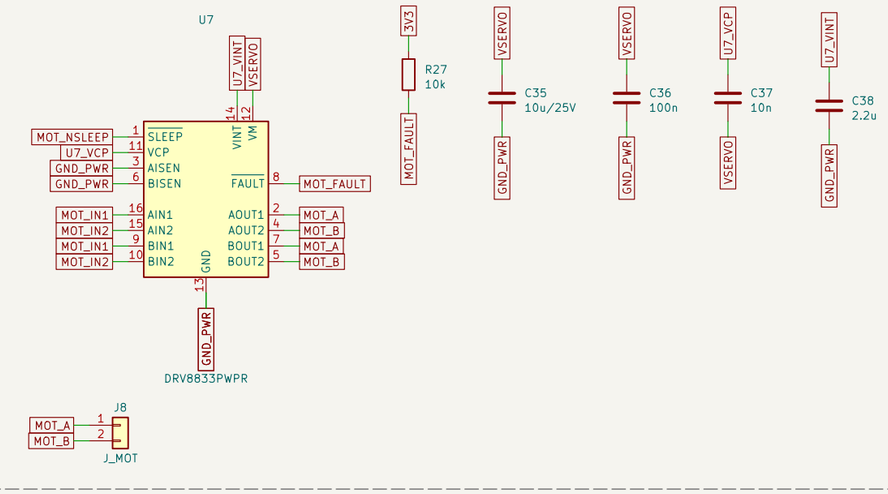
Size: A3	Date: 2026-10-21	**Rev: B**
KiCad E.D.A. 10.0.6		Id: 5/8

Vier Servos und Rollmotor DRV8833 am 2S-Akku

Servos



Rollmotor DRV8833



Power-Flags fuer den ERC

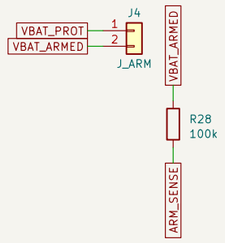


erzeugt aus circuit.py (gen_schematic.py)
Vier Servos und Rollmotor DRV8833 am 2S-Akku
Kartonaut

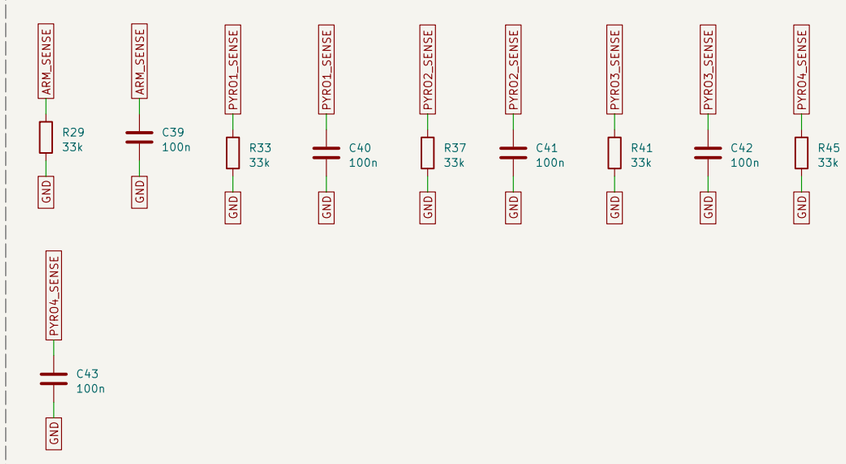
Sheet: /Aktoren/		
File: actuators.kicad_sch		
Title: Mainboard rev B – Aktoren		
Size: A3	Date: 2026-10-21	**Rev: B**
KiCad E.D.A. 10.0.6		Id: 6/8

Vier Pyro-Kanaele low-side (3 und 4 DNP), Arming, Durchgangspruefung

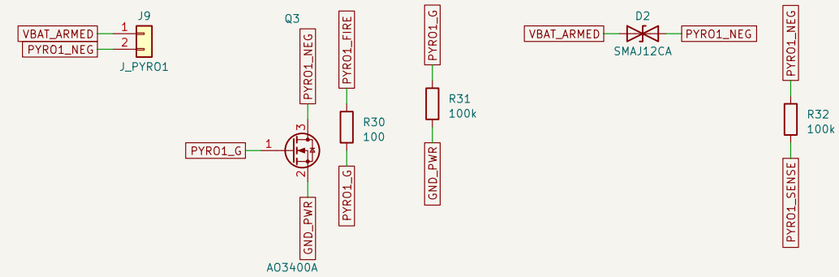
Arming



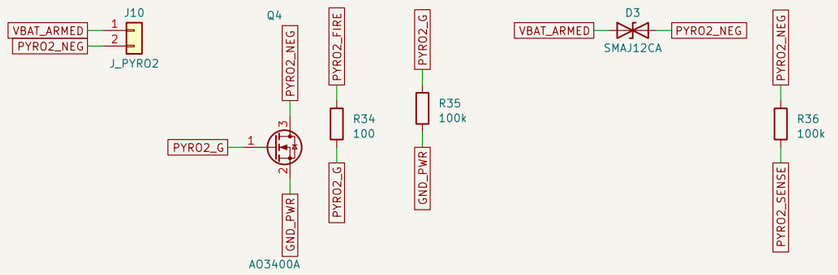
Messteiler am Analogmux



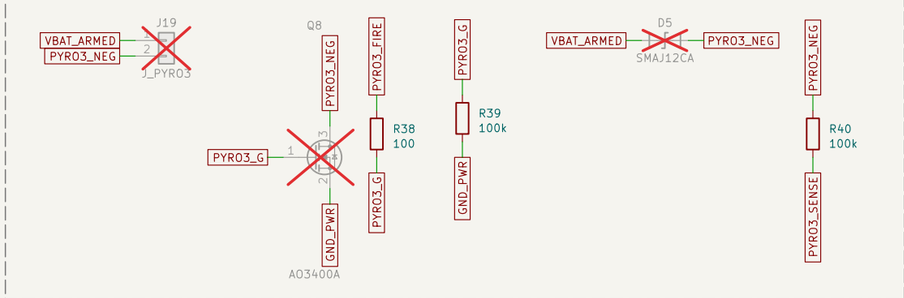
Pyro-Kanal 1



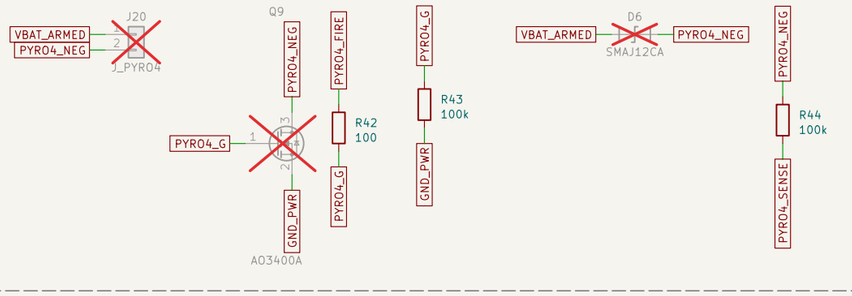
Pyro-Kanal 2



Pyro-Kanal 3



Pyro-Kanal 4



erzeugt aus circuit.py (gen_schematic.py)
Vier Pyro-Kanaele low-side (3 und 4 DNP), Arming, Durchgangspruefung
Kartonaut

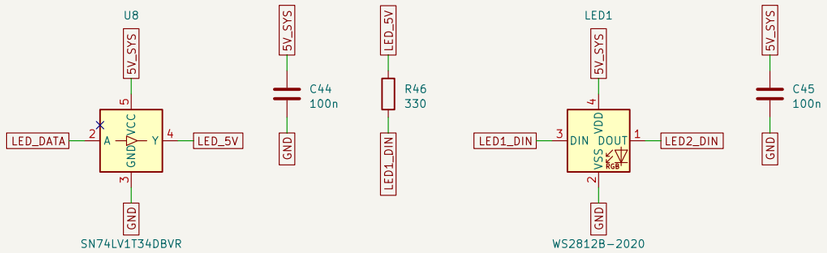
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Title: Mainboard rev B – Pyro

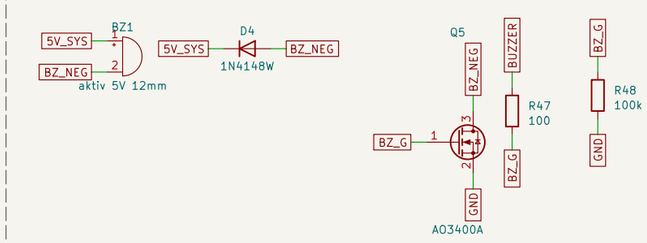
Size: A3	Date: 2026-10-21	**Rev: B**
KiCad E.D.A. 10.0.6		Id: 7/8

WS2812B–Kette mit Pegelwandler, Summer, Kamera–Trigger

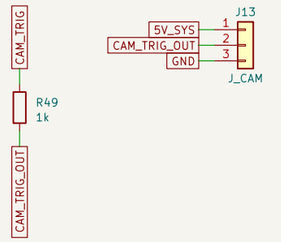
Status–LEDs



Summer



Kamera



erzeugt aus circuit.py (gen_schematic.py)
WS2812B–Kette mit Pegelwandler, Summer, Kamera–Trigger
Kartonaut

Sheet: /Signal/
File: signal.kicad_sch

Title: Mainboard rev B – Signal

Size: A3	Date: 2026–10–21	**Rev: B**
KiCad E.D.A. 10.0.6		Id: 8/8